

Gaming ISO 20022 Payment Messages: A Temporal Adversarial Benchmark for Purpose Codes, Party Fields, and Message Paths

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Abstract—ISO 20022 gives payment controls structured data, but message creators still decide where names, addresses, and purpose information appear. We study *message gaming*: deliberate edits to purpose codes, party fields, or the message path that lower a modelled alert probability while preserving the payment’s economic meaning. A seeded generator gives each debtor–creditor pair latent behaviour and emits pain.001 and pacs.008 records in time order with a controlled rate of ordinary defects. A cost-bounded evader applies up to two of eleven manipulations. Every message is schema-valid and every edit preserves its stated invariants. Every evaluation is temporal: the earliest 60% of a run trains, the next 20% freezes alert thresholds, and the latest 20% is scored, with pair-clustered bootstrap intervals over ten seeds. Screening-evasion lift (SEL) is measured as a paired counterfactual on each manipulated payment. At a frozen 1% alert threshold, where at most 22.4% of gamed messages can be recovered, the best supervised model recovers 18.0% and falls from AUC 0.633 to 0.490 on a withheld misfielding family; the strongest label-free message-level detector recovers 3.3%. A pre-registered test finds that neither the P1/P2/P3 family ($R^2 = 0.35$, exact $p = 0.125$) nor whether an edit leaves an internal contradiction ($R^2 = 0.04$, $p = 0.567$) explains which primitives that model recovers. What separates the families is the evidence required: on withheld path and structure edits every message-level detector is at or below chance, while a history-only control reaches AUC 0.851. Splitting a payment raises per-message lift to 1.44 but lowers attempt-level lift to 0.94 once every child is screened. Results are synthetic and relative to a published screening abstraction.

Index Terms—ISO 20022, payment data quality, purpose codes, sanctions screening, anti-money laundering, adversarial evaluation, synthetic benchmark

I. INTRODUCTION

A payment message is both an instruction and a compliance record. Names, addresses, amounts, and purpose information feed sanctions, fraud, and anti-money-laundering controls, and ISO 20022 makes that information easier to parse. Structure does not remove discretion, though. A message creator still chooses which optional fields to populate, which purpose code to declare, how much detail to place in free text, and sometimes which path a payment takes. Screening is selective by design: Swift’s guiding principles identify the elements that normally warrant direct screening and let other fields provide context [1]. Market-practice guidance records that purpose codes are often misunderstood or placed in the wrong field [2]. The structured-address requirement that CBPR+ had scheduled for 14 November 2026 was deferred by Swift in August

2026 after a community request, with revised timing under consultation [3]. CPMI’s harmonised requirements remain guidance rather than a mandate [4]. Unstructured and half-structured data will therefore coexist with structured data for years, and an ordinary data-quality error and a deliberate low-scrutiny choice can leave the same evidence.

That overlap creates an evaluation problem. A detector can look good by finding malformed records, even though malformed records are mostly benign. It can earn a high AUC by learning the edit signatures of one generator, then fail on a manipulation absent from training. And it can be evaluated with future information leaking into its history features. The question that matters operationally is narrower: at an alert capacity a payment operation could actually run, with thresholds frozen before the scored period, do deliberate changes remain distinguishable from routine error?

The contributions are eleven message-level manipulations with actors, costs, and stated invariants, all schema-valid; a generator with pair behaviour and time order; a temporal protocol with frozen thresholds and pair-clustered uncertainty; paired counterfactual SEL, which needs no separate control group; a pre-registered test of what predicts detectability; and a history-only positive control. Detectors are conventional on purpose. No real payment, customer, or institution data is used.

II. RELATED WORK AND GAP

Public AML resources model who pays whom, how much, and when: the Elliptic data set [5], AMLSim [6], AMLworld [7], SAML-D [8], and the recent AMLgentex and Tide generators with partial observability and configurable prevalence [9], [10]. None encodes the internal placement of ISO 20022 fields, so none can express a manipulation whose mechanism is that a name, code, or narrative sits in a different element. Adversarial machine learning on tabular financial data is the closest methodological line: imperceptible tabular perturbations [11], attacks on imbalanced fraud data [12], and constrained attacks and their benchmark [13], [14]. Those attacks move numeric features under a norm or constraint set; the edits here are discrete, semantics-preserving message operations whose cost is operational, in the classical cost-bounded evasion setting [15], [16]. Noisy payment-address parsing in real traffic [17] is the ordinary-error half of the same overlap. Transaction-purpose

TABLE I
THE ELEVEN PRIMITIVES WITH ACTOR, COST $c(m)$, AND THE
PRE-REGISTERED CONTRADICTION LABEL (C: THE EDIT MAKES TWO
ELEMENTS OF THE MESSAGE, OR AN ELEMENT AND THE CREDITOR’S
PROFILE, INCONSISTENT).

ID	Operation (actor: originator or PSP unless stated)	c	C
P1a	move the ultimate-party name into remittance text	0.30	
P1b	true town and country into a free-text line; TwnNm set to another town	0.25	✓
P1c	truncate the listed party’s name to four characters	0.15	
P1d	omit the ultimate debtor	0.20	
P2a	replace the purpose code by the lowest-weight plausible code	0.05	✓
P2b	replace the category purpose	0.08	✓
P2c	omit the purpose code	0.04	
P2d	declare a salary purpose over a goods or invoice narrative	0.10	✓
P3a	same single payment on the other path (actor with route choice)	0.45	
P3b	multiply the file’s child count by 3 to 7, children sum to the amount	0.35	
P3c	an intermediary replaces the ultimate-party element	0.55	

classification [18] enriches naturally occurring labels rather than detecting strategically chosen ones. No prior benchmark combines message-level field operations with a controlled accidental-error floor and a temporal, capacity-constrained evaluation.

III. THREAT MODEL AND MANIPULATIONS

The adversary can construct or amend a payment instruction but controls neither the screening system nor its threshold. A black-box adversary knows public guidance and receives a coarse accept-or-hold outcome; its admissible set excludes the targeted code substitutions P2a, P2b, and P2d. A grey-box adversary also knows the relative ordering of the benchmark’s purpose-code weights. Route choice (P3a) is a per-record capability held by a PSP, bank-side actor, or multi-channel originator, drawn for 30% of illicit records. Let x be an illicit instruction, m an admissible modification with operational cost $c(m)$, and $\pi(x)$ the alert probability in the published screening abstraction. The evader chooses $m^* = \arg \min_m \{\pi(m(x)) + \lambda c(m)\}$ subject to $V(m(x))$ and $|m| \leq k$, sampling among improving subsets with a softmax of temperature T . V is checked on every record. Amount, currency, debtor and creditor countries, and the economic originator and beneficiary are unchanged by every primitive. P3a changes the path of a single-transaction payment, P3b its decomposition into children that sum to the amount, and P3c the represented party chain, where an intermediary replaces the ultimate-party element. Primitives that need an element the record lacks are inadmissible for that record.

Table I lists the primitives. Whether the P1/P2/P3 family or the contradiction label predicts detectability was fixed as a question, with labels and tests, before the first full-scale run (Section V-D).

IV. BENCHMARK AND PROTOCOL

A. Generator

A run draws $n/20$ debtor–creditor pairs with volume weights from a Dirichlet distribution. Each pair has a creditor industry and a purpose profile drawn around the industry’s profile, a preferred path (pain.001 versus pacs.008), and a batching habit with a typical file size. It also has a party-chain pattern (the

ultimate names it uses and how often one appears), an amount scale, and fixed names and addresses. Messages are emitted in time order by sampling pairs by weight; each carries its index as timestamp. A fraction η receives one ordinary defect (purpose text in remittance, partial address destructuring, absent ultimate party, absent category purpose, in the mix 35/30/20/15%). A fraction ρ is illicit: the listed party is the ultimate debtor or the debtor, and the evader edits the record. Benign remittance text and party names have variable length, so no edit leaves a length that benign traffic does not show; an audit checks this on every run. Records serialise to pain.001.001.09 or pacs.008.001.08. Every instance in the seed-7 test window, every gamed record, and every original was validated against the official XSDs (100.0% of 56,241 valid), and every gamed record passed its invariant check (0 violations). Full distributions are in the supplement.

B. Screening abstraction

A name leg matches listed names against the elements the current path screens: debtor, creditor, and ultimate-party names on pain.001, debtor and creditor names only on the interbank leg. For an illicit record whose listed name sits in a screened element it alerts with probability P_{name} , reduced to $0.39P_{\text{name}}$ when the name is truncated. A score leg combines purpose weight w_p (0.08 to 0.45), corridor weight w_c (0.05 to 0.40), amount, and completeness,

$$s = w_p + w_c + 0.2 \min(1, a/(n_{\text{ch}} \cdot 250,000)) + 0.10u + 0.08t, \quad (1)$$

where a is the payment amount, n_{ch} the number of children, u indicates a missing ultimate debtor and t a missing town or country; only the amount term is diluted by splitting. The leg alerts with probability $p_{\text{score}} = 1/(1 + e^{-8(s-\tau)})$. A residual floor $f = 0.10$ represents controls outside the evader’s reach. Per message, $\pi = 1 - (1 - p_{\text{name}})(1 - p_{\text{score}})(1 - f)$; per attempt, the name leg and floor act once and the score leg once per child, so $\pi_{\text{att}} = 1 - (1 - p_{\text{name}})(1 - f)(1 - p_{\text{score}})^{n_{\text{ch}}}$. The evader optimises the per-message form. Baseline survival is split between the legs by a name share σ ; the main setting $\pi_0 = 0.45$, $\sigma = 0.6$ solves to $P_{\text{name}} = 0.337$, $\tau = 0.782$. Weights are illustrative constants, not any institution’s rule set.

C. Features, detectors, protocol

Message-level features are the consistency checks C1 (code not in list), C2 (town or country missing, address token in free text), C3 (purpose implausible for the creditor’s industry, ultimate debtor absent), and C4 (salary code over a goods narrative). Completeness and length auxiliaries complete the vector. History features use only the pair’s strictly earlier messages: purpose novelty (C5), path novelty, split-count deviation, ultimate-party novelty and presence deviation, and amount deviation. No label enters any feature. D_1 is the fixed rule $C1 + 0.5C2 + C3 + 1.5C4$. D_2 is a gradient-boosted classifier and D_6 a logistic regression on one-hot codes, corridor, amount, and completeness. Both train on gaming labels. D_5 is a local outlier factor on the message-level features of benign training rows [19]. D_3 (isolation forest) and D_4

TABLE II

TEST-WINDOW RESULTS, MEAN $\pm$ BETWEEN-SEED S.D. OVER TEN RUNS OF 200,000 RECORDS. REC. AND PREC. ARE AT THE THRESHOLD FROZEN ON THE VALIDATION WINDOW AT 1% CAPACITY; THE WIDEST PAIR-CLUSTERED 95% HALF-INTERVAL FOR REC. IS 0.020. [†] TRAINS ON GAMING LABELS.

Detector	AUC	AP	Rec.@1%	Prec.@1%	Alerts/hit	Rec.@0.1%
D_1 rules	0.771 $\pm$ 0.004	0.321	0.112 $\pm$ 0.005	0.499	2.0	0.022
D_5 LOF	0.775 $\pm$ 0.005	0.249	0.033 $\pm$ 0.005	0.136	7.5	0.012
D_7 history	0.717 $\pm$ 0.006	0.123	0.035 $\pm$ 0.004	0.205	4.9	0.003
D_2 GBM [†]	0.886 $\pm$ 0.008	0.498	0.180 $\pm$ 0.014	0.818	1.2	0.021
D_6 logistic [†]	0.862 $\pm$ 0.009	0.436	0.155 $\pm$ 0.013	0.712	1.4	0.020
Capacity ceiling	–	–	0.224	1.000	–	0.022

(rules plus forest) are in the supplement. D_7 is the history-only positive control, the standardised sum of the six history deviations. No hyperparameter is tuned on the benchmark.

Each run has $n = 200,000$ records. The earliest 60% trains every learned detector. On the next 20% the score threshold for each alert capacity (1% and 0.1%) is frozen, with a label-blind fraction for tied scores. The latest 20% is the test window on which every number below is computed. Gamed records are 4.5% of the test window, so no detector can recover more than 22.4% at 1% capacity. We report AUC, average precision (AP), and recovery and precision at the frozen threshold, as means over ten seeds with between-seed standard deviations, and pair-clustered bootstrap intervals within each test window. SEL is paired. Each gamed test record contributes its original x and edited $m(x)$, both scored by the same detector with the same prior history, and $\text{SEL} = \sum(1 - \pi_m) / \sum(1 - \pi_0)$. Residual SEL multiplies each term by one minus the frozen alert decision. Leave-one-family-out retrains D_2 and D_6 without one family (at $k = 1$, pure-family records).

V. RESULTS

A. Recovery at a frozen threshold

Table II separates ranking from operational recovery under a threshold frozen before the scored period. Gamed records are 4.5% of the test window, so 1% capacity can recover at most 22.4%. The fixed rules D_1 recover 11.2% at precision 0.499: one alert in two is a gamed message. The local outlier factor D_5 has the best label-free AUC (0.775 against 0.771 for D_1) and the second-best AP, yet the lowest label-free recovery, 3.3%; the history-only control D_7 recovers 3.5%. Among label-free detectors neither AUC nor AP orders recovery at the frozen threshold. The supervised D_2 reaches AUC 0.886 and recovers 18.0% (80% of the ceiling, precision 0.818); logistic regression on the same inputs recovers 15.5%. Realised alert rates on the test window stay within 0.002 of the 1% target, so the frozen thresholds transferred across the window boundary. The exact top-1% ranking metric differs from frozen-threshold recovery by at most 0.011 (supplement). Between-seed standard deviations are at most 0.009 for AUC and 0.014 for recovery.

B. Generalisation, history, and shift

Table III (left) retrains the supervised models without one family and scores pure-family test records. With P1 withheld,

TABLE III

LEFT: LEAVE-ONE-FAMILY-OUT AUC ON PURE-FAMILY TEST RECORDS ($k = 1$, TEN SEEDS; IN $\rightarrow$ OUT FOR THE RETRAINED SUPERVISED MODELS, SINGLE VALUE FOR LABEL-FREE ONES). RIGHT: GENERATOR SHIFT, DETECTORS FITTED AND THRESHOLDS FROZEN ON CONFIGURATION A, SCORED ON B'S TEST WINDOW (THREE SEEDS); B $\rightarrow$ B IN PARENTHESES.

	P1	P2	P3	AUC A $\rightarrow$ B	Rec.@1%	Alert rate
D_1	0.551	0.825	0.383	0.730 (0.730)	0.139	0.017
D_5	0.514	0.834	0.538	0.765 (0.786)	0.019	0.006
D_7	0.576	0.715	0.851	0.697 (0.702)	0.035	0.007
D_2	0.633 $\rightarrow$ 0.490	0.976 $\rightarrow$ 0.641	0.550 $\rightarrow$ 0.502	0.875 (0.897)	0.375	0.045
D_6	0.591 $\rightarrow$ 0.482	0.964 $\rightarrow$ 0.677	0.543 $\rightarrow$ 0.525	0.869 (0.879)	0.336	0.034

D_2 falls from 0.633 to 0.490 (between-seed s.d. 0.018), and no label-free detector exceeds 0.598 on P1, whose records are dominated by name truncation. The inversion behind the low supervised score is systematic. In the reduced training window an absent ultimate debtor is more common among benign records (27%) than among the remaining gamed ones, and the gamed rate given absence is 1.1% against 2.6% given presence. The two P1 edits that remove the ultimate debtor therefore score 0.222 (P1a) and 0.210 (P1d) against benign traffic, while truncation, which leaves it in place, scores 0.568. With P2 withheld the supervised drop is 0.976 $\rightarrow$ 0.641 while the label-free D_1 and D_5 stay at 0.825 and 0.834. P3 is the pre-registered contrast. Every message-level detector is at or below chance (0.383 to 0.538; the withheld D_2 0.502), and the history-only D_7 reaches 0.851. Path substitution, decomposition, and party-chain changes leave the message coherent but break the pair's prior pattern. Table III (right) fits the detectors and freezes thresholds on configuration A, then scores configuration B, whose pair behaviour, industry and corridor mix, defect composition, batching, and string lengths all differ. Rankings transfer: every AUC is within 0.022 of its B $\rightarrow$ B value. Thresholds do not: D_2 's A-frozen threshold alerts 4.5% of B's test window instead of 1%, which is why its recovery rises to 37.5% at precision 0.365, while D_5 's alerts 0.6% and recovers 1.9%. A capacity budget therefore needs recalibration on the target traffic even when the ranking model does not. At 0.5% prevalence (supplement), precision at the frozen threshold falls to 0.080 for D_2 and 0.114 for D_1 , so the fixed rules overtake the supervised model, at 13 and 9 alerts per detected manipulation.

C. Paired screening-evasion lift

Fig. 1 gives paired SEL per primitive. Manipulation raises modelled pass-through by 27–59% per primitive and 40% overall, with pair-clustered half-intervals of at most 0.122. Residual lift shows what each detector removes at its frozen threshold. D_2 pushes the code–narrative mismatch P2d to 0.75 and the omitted code P2c to 0.85, both below 1. It does so by alerting 48.3% and 32.6% of them while alerting under 1% of their unmanipulated originals. The fixed rules bring P2d to 0.96. The path and party-chain edits keep most of their lift under every detector: the smallest residual lifts for P3a and P3c are 1.43 and 1.43. The history-only D_7 is the detector that reaches P3c (10.5% against 0.8% for D_1). Splitting shows

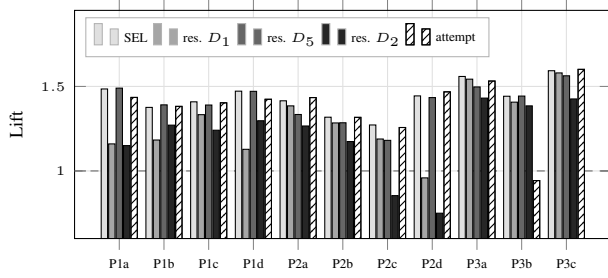


Fig. 1. Paired SEL per primitive on the test window (ten seeds): modelled lift, residual lift after each detector at the frozen 1% threshold, and attempt-level lift when every child transaction is screened.

the value of the attempt-level view. P3b’s per-message lift is 1.44, but once every child transaction is screened its attempt-level lift is 0.94, below 1, so file decomposition pays only against per-message thresholds. For every other primitive the two forms agree within 0.05.

D. What predicts detectability

The reviewed version of this work read its results as showing that detectability follows whether an edit leaves a checkable contradiction rather than its family. We fixed that reading as a hypothesis, with labels and tests, before the first run of the corrected protocol (Table I; supplement). It does not hold. For the supervised D_2 the contradiction label explains $R^2 = 0.04$ of the variance in per-primitive recovery at the frozen threshold (exact permutation $p = 0.567$ over all 330 relabellings; contradiction-bearing primitives 19.8% against 14.8% for coherent ones). The family explains $R^2 = 0.35$ ($p = 0.125$ over 11,550). For the label-free D_5 the figures are 0.00 ($p = 0.955$) and 0.49 ($p = 0.066$). Cost correlates negatively with recovery ($\rho = -0.60$, $p = 0.06$); frequency does not. Leave-one-primitive-out prediction from either grouping is no better than the grand mean (MAE 0.123 and 0.099 against 0.107). The per-primitive rates show why. The omitted purpose code P2c, labelled coherent, is the second most recovered primitive (32.6%), because absence is itself a feature. The address contradiction P1b, labelled contradiction-bearing, is barely recovered (8.6%), because no feature compares the structured town with the free-text line. Detectability follows the checks a detector actually has; a contradiction that no feature reads goes unseen. What does separate the families is the kind of evidence required, as the leave-one-family-out result shows: P3 edits are caught by prior pair history and by nothing message-level. Of the two directional expectations recorded before the rerun, one held: P1 and P3 behave alike for message-level detectors when withheld. One did not: P2 is the easiest family for fixed rules, not the hardest.

VI. DISCUSSION, LIMITATIONS, AND CONCLUSION

Three results survive every protocol correction. Ranking quality does not translate into recovery at a frozen threshold: label-free detectors with AUC between 0.717 and 0.775 recover between 3.3% and 11.2% of gamed messages when 1% of

traffic may be alerted, against a ceiling of 22.4%. The best-ranking one recovers the least. A supervised model that recovers 18.0% in-distribution falls to chance on a family it has not seen, and the reason can be read off the training distribution: the ordinary-error floor makes the signature of the unseen edit look benign. The primitives that reroute, decompose, or re-chain a payment are unseen by message-level checks and visible to prior pair history. The control that addresses them is longitudinal, not a better validator. The claim drawn from the earlier evaluation, that an internal contradiction predicts detectability, did not survive its pre-registered test; what predicts detectability is whether a detector reads the field the edit touches. Under a generator shift the rankings hold while frozen thresholds do not, and at low prevalence precision collapses for every detector.

Limitations. The screening abstraction is transparent, not a reconstruction of any engine, so SEL is a stress measure relative to it. Its weights, the defect mix, pair behaviour, and illicit prevalence are assumptions. The SEL range across the calibration grid (1.14 to 2.85) shows how much the modelled threat depends on them. Generator and detectors share a feature vocabulary; the audit finds 0.0% of benign and 1.2% of gamed test records outside the benign training ranges and at most 5% of any detector’s alert mass on them. Domain ratings of the primitives have not been collected; the questionnaire and blinded procedure are released and the pre-registered test reruns unchanged on raters’ labels. XSD validity is not scheme or business acceptance, and time order is synthetic and without drift.

Artifact

The generator, harness, result files, and verification scripts are public at https://github.com/abhisheksharma2411/ISO-20022_purpose-code_gaming_benchmark (MIT) and archived at <https://doi.org/10.5281/zenodo.23049366>. Every result file records the commit that produced it (17bb868dd2b1). One command regenerates every table and figure, and one command validates every message against the XSDs and every edit against its invariants.

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